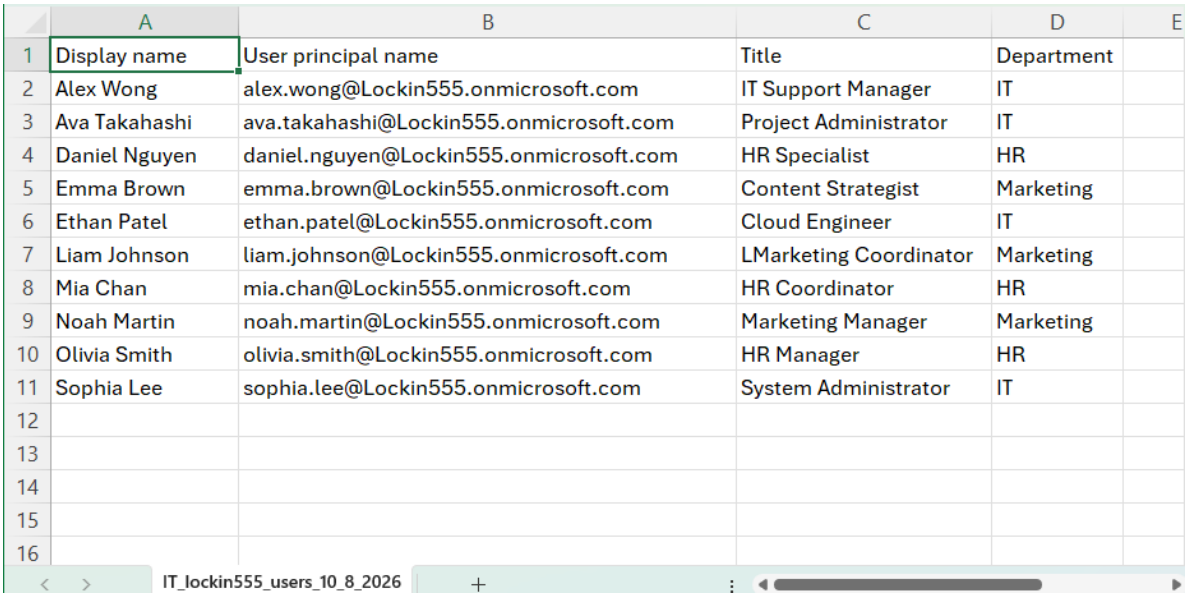
Automated Position Title Comparison: IT Master Data vs. HR Data

IT and HR maintain separate employee data, and discrepancies can occur between the two datasets. In this example, the IT user list is treated as the master data, and the HR user list needs to be checked against it.

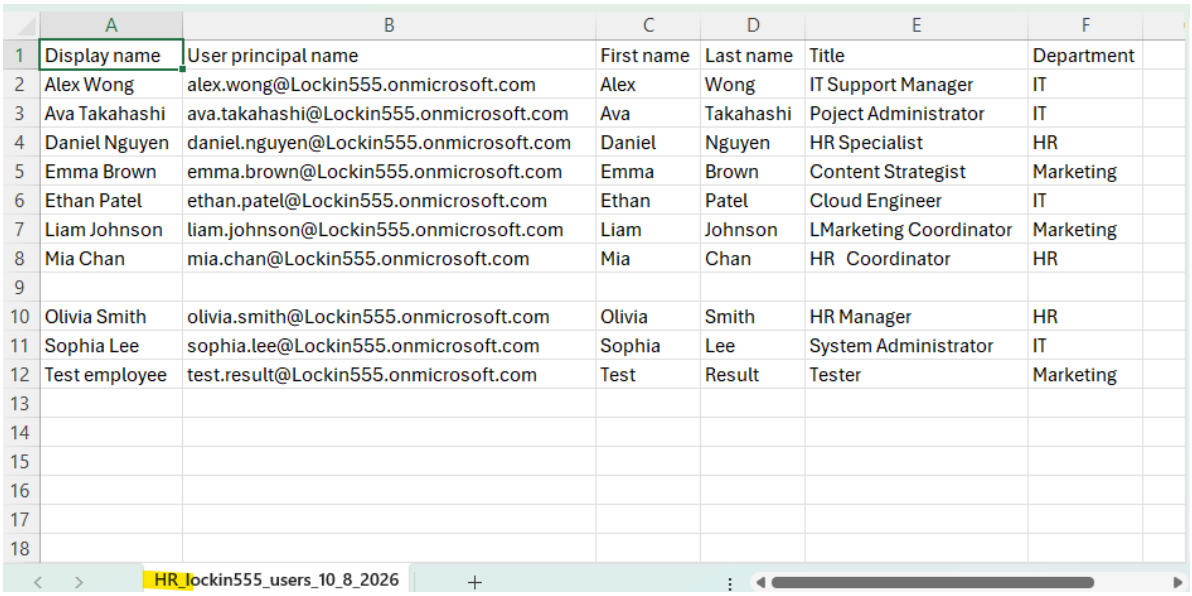
The goal is to automate the comparison of employee position titles using the user's email address as the unique identifier, identify discrepancies, and provide HR with a clear report for review without automatically modifying the original data.



	A	B	C	D	E
1	Display name	User principal name	Title	Department	
2	Alex Wong	alex.wong@Lockin555.onmicrosoft.com	IT Support Manager	IT	
3	Ava Takahashi	ava.takahashi@Lockin555.onmicrosoft.com	Project Administrator	IT	
4	Daniel Nguyen	daniel.nguyen@Lockin555.onmicrosoft.com	HR Specialist	HR	
5	Emma Brown	emma.brown@Lockin555.onmicrosoft.com	Content Strategist	Marketing	
6	Ethan Patel	ethan.patel@Lockin555.onmicrosoft.com	Cloud Engineer	IT	
7	Liam Johnson	liam.johnson@Lockin555.onmicrosoft.com	LMarketing Coordinator	Marketing	
8	Mia Chan	mia.chan@Lockin555.onmicrosoft.com	HR Coordinator	HR	
9	Noah Martin	noah.martin@Lockin555.onmicrosoft.com	Marketing Manager	Marketing	
10	Olivia Smith	olivia.smith@Lockin555.onmicrosoft.com	HR Manager	HR	
11	Sophia Lee	sophia.lee@Lockin555.onmicrosoft.com	System Administrator	IT	
12					
13					
14					
15					
16					

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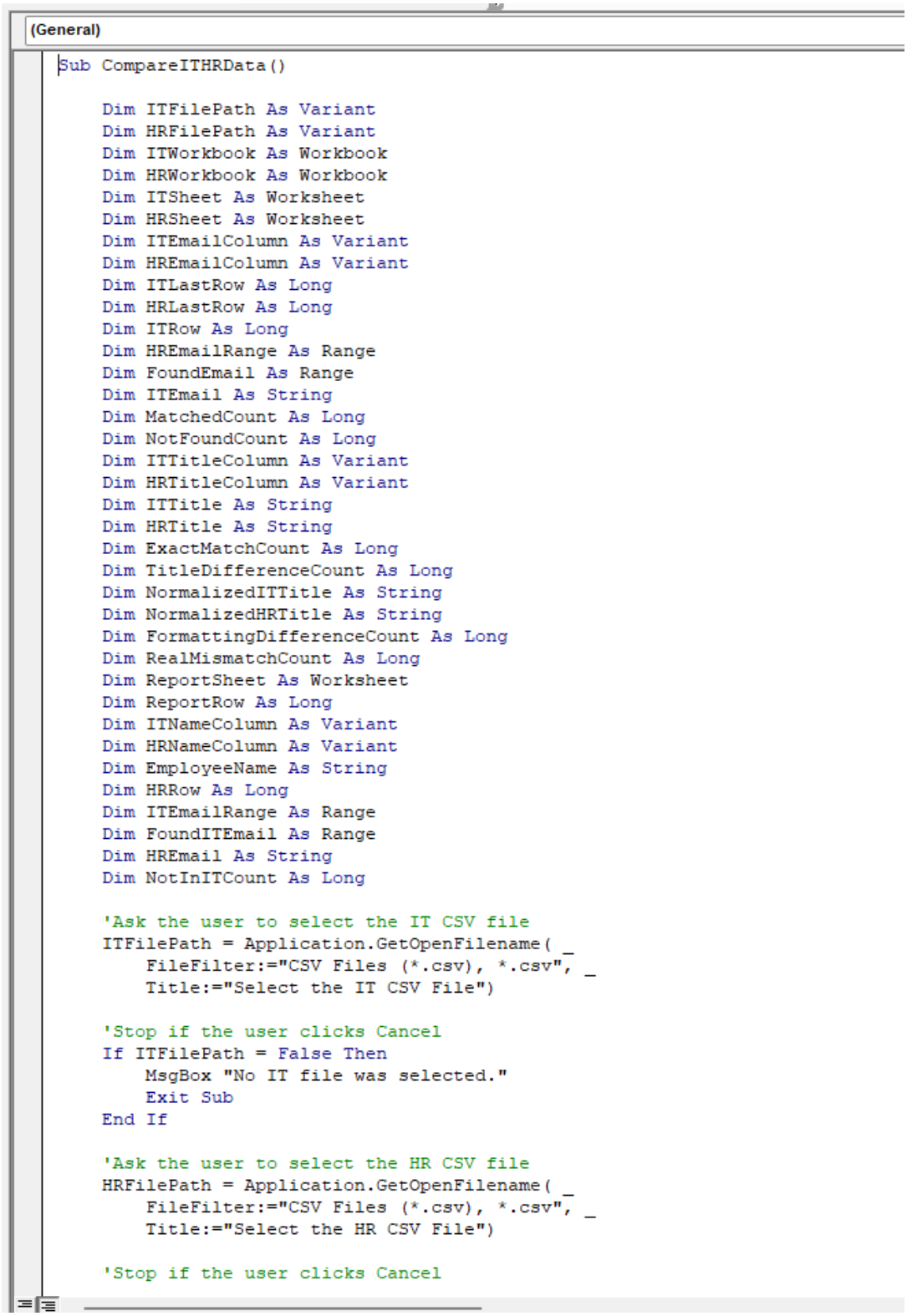
IT user list used as the master dataset for comparison.



	A	B	C	D	E	F
1	Display name	User principal name	First name	Last name	Title	Department
2	Alex Wong	alex.wong@Lockin555.onmicrosoft.com	Alex	Wong	IT Support Manager	IT
3	Ava Takahashi	ava.takahashi@Lockin555.onmicrosoft.com	Ava	Takahashi	Poject Administrator	IT
4	Daniel Nguyen	daniel.nguyen@Lockin555.onmicrosoft.com	Daniel	Nguyen	HR Specialist	HR
5	Emma Brown	emma.brown@Lockin555.onmicrosoft.com	Emma	Brown	Content Strategist	Marketing
6	Ethan Patel	ethan.patel@Lockin555.onmicrosoft.com	Ethan	Patel	Cloud Engineer	IT
7	Liam Johnson	liam.johnson@Lockin555.onmicrosoft.com	Liam	Johnson	LMarketing Coordinator	Marketing
8	Mia Chan	mia.chan@Lockin555.onmicrosoft.com	Mia	Chan	HR Coordinator	HR
9						
10	Olivia Smith	olivia.smith@Lockin555.onmicrosoft.com	Olivia	Smith	HR Manager	HR
11	Sophia Lee	sophia.lee@Lockin555.onmicrosoft.com	Sophia	Lee	System Administrator	IT
12	Test employee	test.result@Lockin555.onmicrosoft.com	Test	Result	Tester	Marketing
13						
14						
15						
16						
17						
18						

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HR user list to be validated against the IT master data.



```
(General)

Sub CompareITHRData()

    Dim ITFilePath As Variant
    Dim HRFilePath As Variant
    Dim ITWorkbook As Workbook
    Dim HRWorkbook As Workbook
    Dim ITSheet As Worksheet
    Dim HRSheet As Worksheet
    Dim ITEmailColumn As Variant
    Dim HREmailColumn As Variant
    Dim ITLastRow As Long
    Dim HRLastRow As Long
    Dim ITRow As Long
    Dim HREmailRange As Range
    Dim FoundEmail As Range
    Dim ITEmail As String
    Dim MatchedCount As Long
    Dim NotFoundCount As Long
    Dim ITTitleColumn As Variant
    Dim HRTitleColumn As Variant
    Dim ITTitle As String
    Dim HRTitle As String
    Dim ExactMatchCount As Long
    Dim TitleDifferenceCount As Long
    Dim NormalizedITTitle As String
    Dim NormalizedHRTitle As String
    Dim FormattingDifferenceCount As Long
    Dim RealMismatchCount As Long
    Dim ReportSheet As Worksheet
    Dim ReportRow As Long
    Dim ITNameColumn As Variant
    Dim HRNameColumn As Variant
    Dim EmployeeName As String
    Dim HRRow As Long
    Dim ITEmailRange As Range
    Dim FoundITEmail As Range
    Dim HREmail As String
    Dim NotInITCount As Long

    'Ask the user to select the IT CSV file
    ITFilePath = Application.GetOpenFilename( _
        FileFilter:="CSV Files (*.csv), *.csv", _
        Title:="Select the IT CSV File")

    'Stop if the user clicks Cancel
    If ITFilePath = False Then
        MsgBox "No IT file was selected."
        Exit Sub
    End If

    'Ask the user to select the HR CSV file
    HRFilePath = Application.GetOpenFilename( _
        FileFilter:="CSV Files (*.csv), *.csv", _
        Title:="Select the HR CSV File")

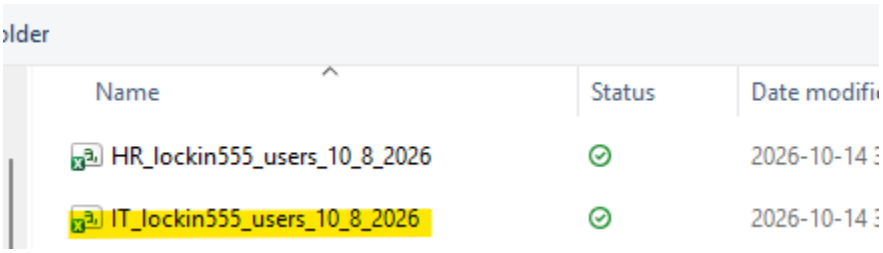
    'Stop if the user clicks Cancel
```

A VBA macro was developed to automate file selection, employee matching, title comparison, and exception reporting.

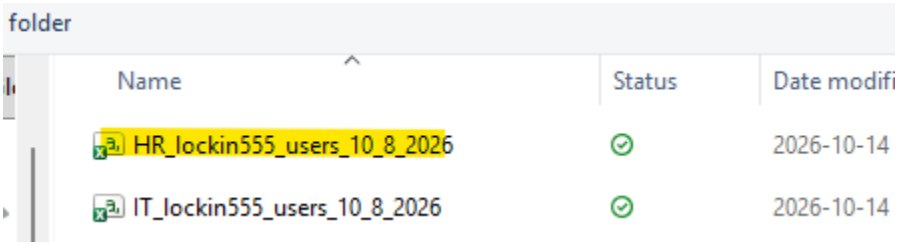


A macro-enabled Excel workbook was created as a simple user interface for running the comparison.

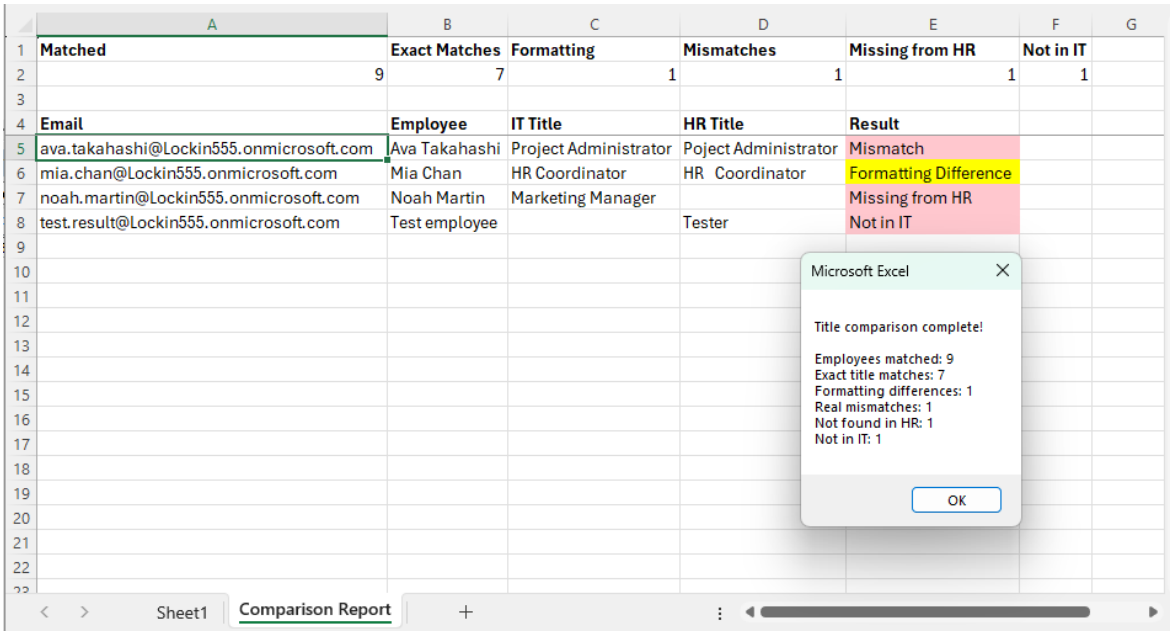
Users can start the comparison by clicking the “Compare IT & HR” button—no VBA knowledge required.



Step 1: Select the IT user list, which serves as the master dataset.



Step 2: Select the HR user list to be compared against the IT master data.



	A	B	C	D	E	F	G
1	**Matched**	**Exact Matches**	**Formatting**	**Mismatches**	**Missing from HR**	**Not in IT**	
2	9	7	1	1	1	1	
3							
4	**Email**	**Employee**	**IT Title**	**HR Title**	**Result**		
5	ava.takahashi@Lockin555.onmicrosoft.com	Ava Takahashi	Project Administrator	Poject Administrator	Mismatch		
6	mia.chan@Lockin555.onmicrosoft.com	Mia Chan	HR Coordinator	HR Coordinator	Formatting Difference		
7	noah.martin@Lockin555.onmicrosoft.com	Noah Martin	Marketing Manager		Missing from HR		
8	test.result@Lockin555.onmicrosoft.com	Test employee		Tester	Not in IT		

Microsoft Excel

Title comparison complete!

Employees matched: 9
Exact title matches: 7
Formatting differences: 1
Real mismatches: 1
Not found in HR: 1
Not in IT: 1

OK

Sheet1 | Comparison Report

After the comparison is complete, a summary pop-up displays the overall results, while the Comparison Report lists only records that require review.

Pink: Data mismatch or user missing from either dataset

Yellow: Formatting difference, such as extra spaces or capitalization

***Exact matches are excluded from the report, allowing users to focus only on exceptions.**

	A	B	C	D	E	F
1	Display name	User principal name	First name	Last name	Title	Department
2	Alex Wong	alex.wong@Lockin555.onmicrosoft.com	Alex	Wong	IT Support Manager	IT
3	Ava Takahashi	ava.takahashi@Lockin555.onmicrosoft.com	Ava	Takahashi	Poject Administrator	IT
4	Daniel Nguyen	daniel.nguyen@Lockin555.onmicrosoft.com	Daniel	Nguyen	HR Specialist	HR
5	Emma Brown	emma.brown@Lockin555.onmicrosoft.com	Emma	Brown	Content Strategist	Marketing
6	Ethan Patel	ethan.patel@Lockin555.onmicrosoft.com	Ethan	Patel	Cloud Engineer	IT
7	Liam Johnson	liam.johnson@Lockin555.onmicrosoft.com	Liam	Johnson	LMarketing Coordinator	Marketing
8	Mia Chan	mia.chan@Lockin555.onmicrosoft.com	Mia	Chan	HR Coordinator	HR
9						
10	Olivia Smith	olivia.smith@Lockin555.onmicrosoft.com	Olivia	Smith	HR Manager	HR
11	Sophia Lee	sophia.lee@Lockin555.onmicrosoft.com	Sophia	Lee	System Administrator	IT
12	Test employee	test.result@Lockin555.onmicrosoft.com	Test	Result	Tester	Marketing

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Detected issues are also highlighted directly in the HR dataset, making it easy for HR to locate and investigate the affected records. **The IT master dataset remains unchanged and is used only as the source of truth for comparison.**

The tool identifies discrepancies without automatically modifying the source data, allowing HR to review each exception and determine whether a correction is required.

The Comparison Report is automatically cleared and regenerated each time the tool is run, ensuring that it always reflects the latest comparison.

Outcome: A reusable Excel VBA tool that turns a manual IT–HR data reconciliation process into a simple, exception-based review workflow.